Deepak Enamala

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.86
FIIT-JEE, Jr. College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2020	8.94%
Ravindra Bharathi School	Class X (Board of Secondary Education Andhra Pradesh)	2018	9.50%

PROJECTS

Retail Sales Forecasting & Business Analytics:

(https://github.com/Deepak14002/retail-sales-forecasting.git)

- Developed a time-series revenue forecasting pipeline using lag, rolling-window, calendar, and seasonal features on the UCI Online Retail dataset.
- Compared Linear Regression, Gradient Boosting, and Random Forest models; tuned Random Forest achieved MAE of $11,051 and RMSE of $15,451, with SQL-based business analysis and a Streamlit dashboard.

Customer Churn Prediction & Business Analytics:

(https://github.com/Deepak14002/customer-churn-analytics.git)

- Built an end-to-end customer churn prediction pipeline on the IBM Telco dataset using Logistic Regression and Gradient Boosting**,** with preprocessing, cross-validation, and hyperparameter tuning.
- Performed customer segmentation and statistical analysis; optimized the classification threshold to improve churn detection, achieving 71.1% recall and 0.617 F1-score at the selected threshold.

ML-Based RIS Beamforming Optimization:

(https://github.com/Deepak14002/ris-ml-beamforming.git)

- Developed an ML-based passive beamforming system for a 16-element RIS, learning discrete 2-bit phase configurations from the cascaded BS-RIS-user channel using a multi-head neural network.
- Compared Random RIS, quantized analytical optimization, Coordinate Descent, and ML-based beamforming; the ML approach achieved 20.90 dB average SNR and 6.955 bits/s/Hz, retaining approximately 99.5% of the analytical reference rate.

SKILLS

Programming Languages: C, C++, Python, Verilog

Technical Skills: Data Structures & Algorithms, Data Analytics, Machine Learning, Deep Learning, SQL

Tools & Software: MATLAB, Quartus, kiCad, EasyEDA

EXPERIENCE

Qmax Test Equipments Pvt. Ltd. -- Industrial Intern

Jun 2023 - Jul 2023

- Created **annotated image datasets** by labeling electronic components across multiple views of PCB images for automated component recognition.
- Supported development of a **computer-vision-based electronic board inspection system** to identify component locations and reduce manual tagging.
- Contributed to automating the component identification stage of **PCB testing and functional inspection**.

RELEVANT COURSES

Random Variables and Stochastic Processes, Matrix Theory, Principles of Digital Communication, Digital Signal Processing, Information Theory and Coding, Pattern Recognition and Machine Learning, Wireless Communication, Advanced Communication Engineering

SCHOLASTIC ACHIEVEMENTS

Published **"Minimizing Congestion in Ring Topology Using Node-to-Node Path-Finding Algorithm"** in the IEEE International Conference on the Advances in Data Engineering and Intelligent Computing Systems ADICS-2024 held on April 17th and 18th 2024. **DOI:** 10.1109/ADICS58448.2024.10533584

IET Prize 2023 -- Received the **IET Student Award for academic distinction** in Electronics and Communication Engineering at Hindustan Institute of Technology and Science.